VENKATESH MALLARAPU

Data Scientist | ML Engineer | MLOps
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PROFESSIONAL SUMMARY

Results-driven Data Scientist and ML Engineer with hands-on expertise in Python, SQL, Machine Learning, and MLOps. Skilled in building end-to-end ML pipelines, CI/CD automation for ML lifecycle using CML workspaces, model governance and lineage, secure REST API model serving, and cloud-based deployment. Projects span healthcare AI, NLP, recommendation systems, and time-series forecasting. Pursuing B.Tech in Computer Science with a strong foundation in statistics, algorithms, and database management.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Regression, Classification, Clustering, Random Forest, SVM, KNN, Naive Bayes, LSTM, XGBoost, Decision Trees
MLOps & CI/CD: CML Workspaces, CI/CD Pipelines for ML Lifecycle (GitHub Actions), Model Versioning, Secure REST API Model Serving
Governance & Monitoring: Model Governance, Lineage Tracking, Drift Detection, Model Monitoring Tools Integration
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Flask, Matplotlib, Seaborn
Tools & Platforms: Jupyter Notebook, Google Colab, Git/GitHub, VS Code
Data Analysis: EDA, Feature Engineering, Data Cleaning, Preprocessing, SMOTE, Outlier Handling
DS & Algorithms: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Maps, Dynamic Programming, BFS/DFS
Gen AI & LLMs: Prompt Engineering, RAG Pipelines, LLM Integration (OpenAI/HuggingFace), Code Generation & Translation
Enterprise Search: Elasticsearch (indexing, querying, relevance tuning), Splunk (log analysis, dashboards, alerts)

EXPERIENCE

Data Scientist Intern · Evoastra Ventures *Feb 2026*

- Performed EDA, data preprocessing, and feature engineering on real-world business datasets to identify key patterns and inform model design.
- Built and evaluated ML models (Random Forest, SVM, Regression) for AI-based supply chain optimization and demand forecasting, improving prediction accuracy by ~15%.
- Deployed models via REST APIs for real-time inference; implemented CI/CD pipeline steps for ML lifecycle automation enabling reproducible, versioned deployments.
- Integrated model monitoring and governance tools to track performance post-deployment; maintained model lineage documentation across experiment iterations.
- Created dashboards and data visualizations using Matplotlib and Seaborn to communicate actionable insights to stakeholders.

PROJECTS

Plant Disease Prediction | TensorFlow/Keras, VGG16, Flask, REST API, OpenCV

- Built a Transfer Learning (VGG16) classifier on the PlantVillage dataset achieving 96%+ accuracy across 38 disease categories; deployed as a Flask web app with a REST API endpoint for real-time inference; implemented model versioning and lineage tracking across training iterations.

Credit Card Fraud Detection | XGBoost, SMOTE, Scikit-learn, CI/CD Pipeline

- Developed fraud detection system achieving precision-recall AUC of 0.95+ on highly imbalanced data using XGBoost and Random Forest with SMOTE oversampling; built a CI/CD pipeline for automated model retraining with governance and lineage tracking; visualized performance using confusion matrices and ROC curves.

Handwritten Character Recognition | CNN, TensorFlow/Keras, OpenCV, REST API

- Trained a CNN on MNIST and EMNIST datasets achieving 98%+ accuracy with image preprocessing via OpenCV; deployed a real-time prediction interface via REST API endpoint; demonstrated the full ML pipeline from data ingestion and augmentation to model training, evaluation, and live inference.

EDUCATION & CERTIFICATIONS

B.Tech Computer Science · Holy Mary Institute of Technology & Science, Hyderabad *May 2026*
Machine Learning · Visualpath **Data Science** · Tutedude